

Sajjad Pakdamansavoji

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SUMMARY

Passionate researcher/engineer with 4+ years of experience in designing and implementing AI solutions for software products. Technical advisor to 4 ML-driven SMEs and startups. Dedicated to delivering bug-free, scalable and well-documented programs.

EXPERIENCE

2024–Now	Senior Computer Vision Researcher, Huawei, Toronto, ON, Canada. - Led the perception team to develop solutions for 6D pose estimation, 3D reconstruction, scene graphs, and SLAM. - Designed and integrated algorithms for robot perception, planning, and natural language understanding.
2023–2024	Technical Advisor & Applied AI Intern, Vector Institute for AI, Toronto, ON, Canada.
May–May	- Led weekly consultation sessions with 4 SMEs on cloud AI integration, model selection, and strategic product design. - Delivered an NLP-focused training workshop on model development and deployment for over 20 startup teams. - Developed an assessment tool leveraging RAG and open-source LLM prompting to automatically generate QAs. - Boosted end-to-end accuracy by 10% through human evaluation and improved RAG's mean reciprocal rank (MRR) by 15%. - Built a CI/CD pipeline on GitHub to seamlessly integrate code and documentation into a fully operational website.
2021–2023	ML Engineer, Trans-Plan, Toronto, ON, Canada.
Spet–May	- Engineered and deployed an ML software stack for multi-target, multi-source object detection, tracking, and segmentation. - Elevated object trajectory classification accuracy to 86% in single-camera setups and 92% in multi-camera configurations. - Designed an interactive GUI with PyQt5 and OpenGL, facilitating visualization, annotation, and human-in-the-loop feedback. - Actively participated in code reviews, documentation, code refactoring, and version control to enhance code quality.
2019–2019	ML Engineer Intern, AVIR Company, Tehran, Iran.
July–Sept	- Developed and deployed a multi-modal ML software to translate Iranian Sign Language (ISL) translation from video to text. - Extracted features from body and face key-points, trained seq-2-seq models for translation and classification. - Elevated classification accuracy from 82% to 90% at the alphabet level and from 60% to 65% at the word level in translation.

PUBLICATIONS

IEEE ITSC 2024	Ground Plane Projection for Improved Traffic Analytics at Intersections; S. Pakdamansavoji, J. Elder, et al..
IEEE WACV 2024	A Multi-Camera/LiDAR, All-Weather, Day/Night Intersection Traffic Dataset; K. Bhargava, S. Pakdamansavoji, et al. (submitted).

PROJECTS

2023–2024	Structured QA with Retrieval-Augmented Generation Vector Institute in AI.
May–May	- Created an automated QA system utilizing GPT-4 and LaMMMA2 to streamline the processing of technical SOW documents. - Designed and developed a robust RAG pipeline and integrated LLMs for structured generation of QAs in JSON format. - Implemented advanced hierarchical summarization techniques, enhancing context accuracy and relevance for LLM inputs. - Collaborated on system integration and testing, achieving a 10% increase in content precision with human evaluation.
2021–2023	3D Multi-Camera Multi-Object Turning Movement Counts Trans-Plan.
Sept–May	- Executed data cleaning, transformation, feature engineering, and camera calibration on a large proprietary dataset. - Architected an open-sourced ML software stack for object detection, tracking, segmentation, pose estimation and classification. - Effectively executed distributed multi-GPU fine-tuning and inference on a vision foundation model (InternImage). - Improved tracking MOTA metric by 28% and enhanced trajectory classification accuracy from 80% to 92%.
2021–2022	Synthetic Face Generation Based on Celebrity Faces AVIR Company.
Dec–May	- Conducted a performance comparison of generative models on face generation tasks using the CelebA public dataset. - Trained a range of models, including GANs, Normalizing Flow (RealNVP), Autoencoders (VAE), and diffusion models. - Ranked models based on Inception Score (top IS≈113.9) and Fréchet Inception Distance (top FID≈8.2).

EDUCATION

2021–2024	MSc in Computer Science, York University, Canada. GPA 8.6/9.
2016–2021	BSc in Electrical & Computer Eng, University of Tehran, Iran.

AWARDS

2022–2023	VISTA Scholarship, Center of Vision Research (20K CAD).
2021–2022	Vector Scholarship in AI, Vector Institute (17.5K CAD).
2021–2022	Lassonde Entrance Scholarship, York University (6K CAD).
2017–2018	Founders Scholarship, University of Tehran.

SKILLS

Programming	Python(advance), C++(advance), C, Java, MATLAB, Bash
Frameworks	Pytorch, Tensorflow, Keras, JAX, GCP, OpenAI
Libraries	OpenCV, OpenMMLab, TensorRT, ONNX, Pillow Scikit-learn, NLTK, HuggingFace, spaCy, Matplotlib Numpy, Pandas, Seaborn, Docker, Git, Jupyter
Soft Skills	Communication, Creativity, Teamwork, Management Continuous Learning, Analytical Thinking, Empathy
Languages	English (fluent), Farsi (native), Arabic (intermediate)